WEB PROGRAMMING EXAM CHEAT SHEET

PAGE 1 – HTML + CSS + JavaScript BASICS

COLUMN 1 – HTML ESSENTIALS

What is HTML

- Markup language for structuring web pages
- Uses *tags* to describe content

Basic Structure

```
<!doctype html>
<html>
  <head>
    <title>Title</title>
  </head>
  <body>
    Content
  </body>
</html>
```

Common Tags

- Text: h1–h6, p, span, br, hr
- Grouping: div, section, article, main, aside, nav
- Lists: ul, ol, li, dl, dt, dd
- Media: img, audio, video
- Links: a href="URL"
- Tables: table, tr, td, th

Global Attributes

- id, class, style, title
- hidden, draggable, contenteditable

Forms

- input, textarea, select, option, button
- Input types: text, password, radio, checkbox, submit

CSS CORE CONCEPTS

CSS Purpose

- Controls layout & appearance
- Separates style from structure

Syntax

selector { property: value; }

Ways to Add CSS

- Inline (highest priority)
- External stylesheet
- Internal <style>

Selectors

- Type: p, div
- Class: .box
- ID: #main
- Attribute: input[type=text]
- Descendant: div p
- Child: div > p

Pseudo-classes

- :hover, :active, :focus, :checked, :first-child

Box Model

- Content → Padding → Border → Margin
- Total size ≠ width/height

Display & Visibility

- display: block | inline | none
- visibility: hidden

CSS LAYOUT & POSITIONING

Positioning

- static (default)
- relative (offset from itself)
- absolute (relative to positioned ancestor)
- fixed (relative to viewport)

Z-Index

- Controls stacking order (higher = front)

Float

- float: left | right
- Content wraps around

Text Properties

- color, font-size, font-family
- text-align, line-height, text-transform

Transitions

transition: all 0.5s;

Responsive Design

```
&lt;meta name="viewport" content="width=device-width, initial-scale=1.0"&gt;
```

JAVASCRIPT + PROMISES + jQUERY

JAVASCRIPT FUNDAMENTALS

JavaScript Basics

- Client-side scripting language
- Dynamically modifies HTML/CSS

Variables

```
let x = 5;
var y = "text";
```

- Dynamically typed
- undefined if not initialized

Control Structures

- if / else, switch
- Loops: for, while, for…of

Functions

```
function add(a,b){ return a+b; }
```

Events

- onclick, onload, onchange, onsubmit

DOM Access

```
document.getElementById("id");
document.querySelector(".class");
```

ASYNC, PROMISES & JSON

Asynchronous Programming

- JS is non-blocking
- Uses callbacks, promises, async/await

Promises

```
new Promise((resolve, reject) => {
  if (ok) resolve(val);
  else reject(err);
});
```

Promise Chain

```
fetch(url)
  .then(r => r.json())
  .catch(err => console.log(err));
```

Async / Await

```
async function load() {
  try {
    let data = await fetch(url);
  } catch(e) {}
}
```

Error Handling

- Promises → .catch()
- Async → try / catch

JSON

- Data exchange format
- JSON.parse() → object
- JSON.stringify() → string

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6 – jQUERY & AJAX

jQuery Basics

- JS library to simplify DOM & AJAX

Syntax

```
$(selector).action();
```

Selectors

- $("#id"), $(".class"), $("p:first")

DOM Manipulation

```
$("div").addClass("box");
$("p").text("Hello");
$("p").append("text");
```

Events

```
$("button").click(() => {});
```

Document Ready

```
$(function() { });
```

Effects

- hide(), show(), fadeIn(), slideUp()

AJAX

```
$.get(url, callback);
$.post(url, data, callback);
```